Unit - III

① Role of Transportation Engineering and economic development of country :-

1. Transportation contributes to the economic, industrial, social & cultural development of country.
2. Transportation is viatle for the economic development of any region since every commodity produced whether it is food and clothing, industrial products (or) medicine needs transport at all stages from production to distribution.
3. In the production stage transportation is required for carrying raw materials like seeds, coal & steel etc. In the distribution state transportation is required from production centers, farms & factories to the marketing centers & later to the retailers and the consumers for distribution
4. The inadequate transportation facilities retard the process of social economic development of the country
5. The adequency of transportation system of a country indicates its economic & social development
6. The economic activities are the process by means of which the products are utilised to satisfy human beings.
7. The 2 important factors well known in economic activity are
   a) production (or) supply
   b) consumption for human wants (or) demands

8. The importance of transportation in economic activity is to be found in its effects and both human wants for goods and satisfaction through production & distribution.

Pavement :- ②

It can be defined as the structure which separates the tyres of vehicles from the under laying foundation

* Pavement is the upper part rail way, road way, air port (or) parking area structures

Different ~~two~~ layers of pavement :-

1. subgrade (3)
2. sub Base
3. Base
4. surface course

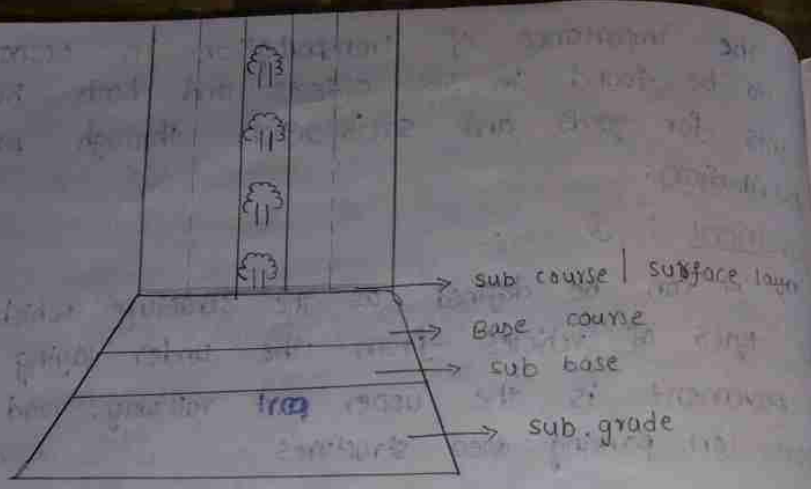
Types of pavements :-

1. Flexible pavement
2. Rigid pavement

Flexible pavement :- ~~F~~

* Flexible pavements are those which are having low or neligable or flexural strength and are rather flexible in their structural action under the loads.
* The flexible pavement layers may reflect the non-~~renewa~~ recoverable as well as recoverable deformation of the pavement surfaces
* Different layers of the flexible pavements are

1. ~~subgrade~~ surface course | surface layer
2. ~~Su~~ Base
3. Sub Base
4. Sub grade



Surface layer:-

* The surface layer is the layer in contact with traffic loads and normally contains the highest quality materials
* It provides characteristics such as friction, smoothness, noise control & resistence against load & drainage

Base course :-

* The base course is immediately beneath the surface course. It provides additional load distribution and contribution to drainage and frost resistence
* Base course are usually constructed by aggregates

Sub Base course :-

* The sub base course is b/w the base course and sub grade. It functions primarily structural supports to the pavement.

Sub Grade :-

* The sub grades are composed of wide range of materials usually consists of hard stratum of soil.

Rigid pavement :-

* Rigid pavements are those which contains sufficient beam strength to be able to bridge over the localised subgrade failures and areas of inadquate supports.

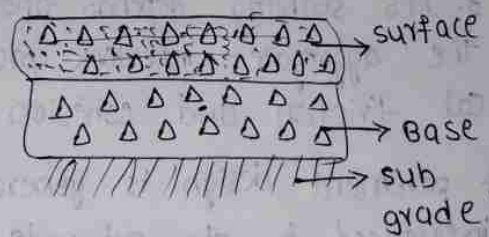
* The load is transmitted through beam action of the slab in rigid pavements.

* The rigid pavements are those which possess note worthy flexural strength or flexural rigidity

* The rigid pavements are generally made of cement concrete therefore it is called as cc pavements.

* In rigid pavements the stresses are not transmitted from grain to grain to the lower layer as in case of flexible pavement layers.

* The components of rigid pavements are
  1. surface course
  2. Base
  3. sub grade



Surface course :-

* The surface course is the layer in contact with traffic loads and is made of PCC (portland cement concrete) or RCC.

* It provides characteristics such as friction, smoothness, noise control & drainage in addition to that it serves as water proofing layer to the underlaying base & sub grade

* The surface course can carry the loads of vehicles the thickness is varies usually b/w 150mm to 300mm 150mm for light loading & 300mm for heavy loading

Base course :-

* The base course immediately beneath the surface course it provides additional load distribution and contributes resistances against frost action and provides drainage

Differences b/w flexible pavement & rigid pavement

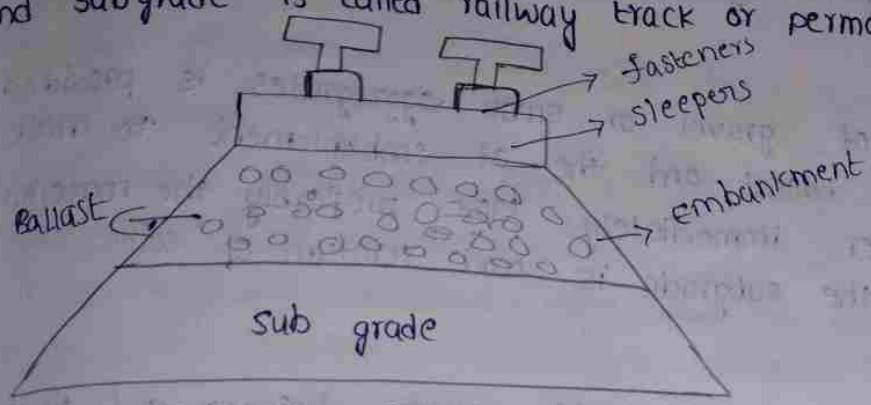
Flexible pavement	rigid pavement
1. It consists of a series of layers with the highest quality materials at are near the surface of pavement.	1. It consist of one layer PCC slab or RCC slab
2. It reflects the deformations of subgrades and subsequent layers on the the surface of the road	2. It is able to bridge over localised failure and area of inadequate support
3. Its stability depends upon the aggregate interlock, practical friction and cohesion	3. The structural stability is provide by the pavement slab itself by its beam action
4. pavement design is generally influenced by the subgrade strength	4. Flexural strength of concrete is a major factor of design
5. It functions by a way of load distribution to the component layers.	5. It distributes load over wide area of subgrade because of its residitity and high modulas of velocity
6. Temperature variation due to change in atmospheric condition do not produce stress in flexible pavement	6. Temperature changes induce heavy stress in the rigid pavements.

Railway Engineering :-

Branch of civil engineering which deals with the design, construction and maintainence of the railway tracks for safe and efficient movements of train is called Railway Engg.

Permanent way (or) Railway track :-

The combination of Rails, fitied and sleepers with the help of fixtures and fastenings and Resing and Ballast and subgrade is called railway track or permanent track.



Components of Railway (or) Tracks

1. Rails
2. Sleeper
3. Ballast
4. sub grade
5. Embankment

Rails :-

The high carbon rolled steel section which are laid end to end in two parallel lines or sleepers to provide continous and level surface for the trains to move carrying acclear rods of wheels of train are called rails.

Sleepers:

Sleepers are the transverse ties that are laid to support the rails. They have an important role in the track as the transmit the weighing loads from rails to the ballast.

Several types of sleepers are used in Indian railways are made by wood PCC or RCC.

3. Ballast :-

* Ballast is a broken stone layer of broken stones and aggregrates or any other granular material placed and tracked below and around sleepers. For distributing the load from sleepers to embankment. (sub grade)
* It provides drainage as well as longutudional and lateral stability to the track.

Sub grade :-

* A layer of gravel or small aggregrates is provided between the ballast and the at embankment to drain of the water immediately. Most preferbly the material available at the subgrade is hard stratum of soil.

Embankment :-

* It is prefered to provide proper drainage and level surface for the track
* It is prepared by levelling the ground and necessary cutting or filling
* The embankment is provided side slope of 1:2

Airport Engineering :-

Airport Engineering deals with the planning, design and construction of terminals, runways and navigation aids to provide for passengers and carbo services.

Airport :-

* An Airport is a facility where passenger connect to ground transportation to Air transportation
* It is a location where air craft such as aeroplanes, helicopters take off and landing

* Air craft may also be stored or maintained at an airport

Types of Airports :-

* International Airports
* Domestic Airports
* Regional Airports

International Airports :-

* An Internation airport has direct service to many other airports
* It handle scheduled commerical airlines both for passengers and carbo
* I.A typically equiped with customs and integration facilities to handle international flights and to from other countries.
* such airports are usually larger and having longer run ways and facilities to accomdate the large air craft

Domestic Airports:-

* The domestic airport is an airport which handle domes flights or flights within the same country.
* Domestic airports does not have customs and integratio facilities and are therefore incapable of handling flights to or from a foreign airport
* These airports normally have short runways which are sufficient to handle short / medium / aircraft.

Regional Airports :-

* A Regional Airport is an airport surving traffic within a relatively small or lightly populated geogra area
* A Regional airport usually does not have customs integration facilities to process traffic b/w coun

* Air craft using these airports tend to be smaller business jets. or private aircraft

Harbours - (3)

A Harbour can be defined as a sheltered area of the sea in which ships could be launch, built or taken for repair or could seek refuse in time of storm or provide for loading and unloading for passenger and carbo

Types of Harbours :-

Harbours are classified as follows:

1. Natural Harbours
2. semi natural Harbours
3. Artifical Harbours

Natural Habours :-

Natural formations affording safe discharge facilities for ships on seacoast are called natural Harbours.

* With the rapid development of navys engaged either in commerce or war improved accomidation and facilities for repairs, storage of carbo and connected Amenities had to be provided in natural harbours.
* The size and draft of present day vesseles have necessated the works improved for natural harbours
* The factors such as local geographical features, growth of population development of the area have made the natural harbours big and attractive

Ex :- Bombay harbour

Semi natural harbour :-

These type of harbour is protected and on side by head lands production and it requires man made production only at the entrance

Ex :- vishakapatnam harbour.

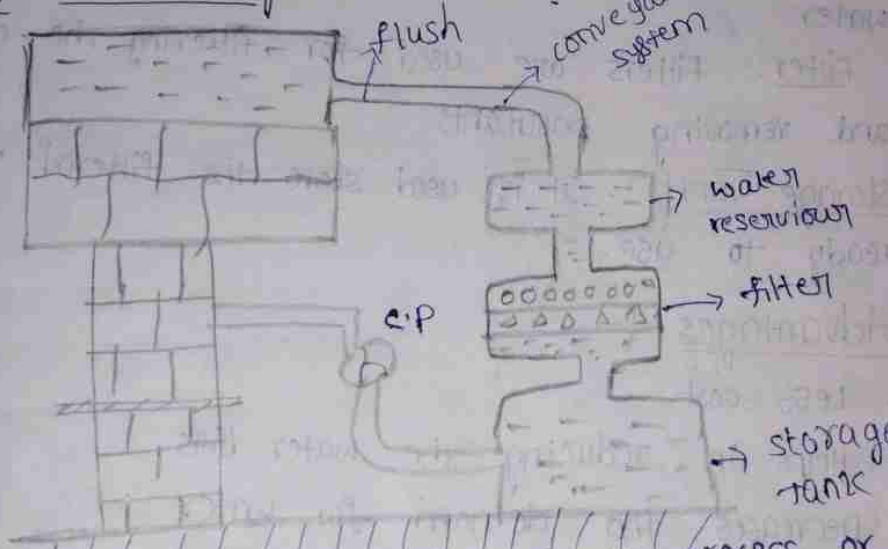
Artifical Harbours :-

where such natural facilities are not available countries having a sea port had to create or construct such shelters making use of engineering skill & methods & such harbours are called ~~Artifical~~ harbours or Man Made harbours

Ex :- Madras harbour

Rain water Harvesting :-

(5)



Rain water Harvesting is the simple process or technology used to conserve rain water by collecting, storing, conveying and purifying of the rain water that runs off from roof tops, parks, roads, & open grounds etc... for later use.

Components of rain water harvesting :-

1. catchment area
2. conveyance system
3. Flush
4. Filter
5. Storage Tanks.

catchment area :-

* It is used to collect and store the captured rain water

Conveyance system :-

* It is used to transport the harvested water from the catchment to the recharge zone

Flush :- It is used to flush out the first spell of rain water.

Filter :- Filters are used for filtering the collected rain water and removing pollutants.

Storage Tanks :- It is used store the filtered water which is ready to use.

Advantages :-

1. Less cost
2. Helps in reducing the water bills
3. Decreases the demand for water
4. Reduces the need for imported water
5. promotes both water & Energy conservation
6. Improves the quality and quantity of ground water
7. This technology is relatively simple. & easy to install & operate

Disadvantages :-

1. Regular maintainence is required
2. Requires some technical skills for installation

3. Limited & no rainfall can limit the supply of rain water
4. If not installed correctly it may attract mosquotes & other water borne diseases

Sources of water :-

All the source of water can be broadly divided into 1. surface sources
2. sub surface sources

The surface sources are further divided into
(i) streams
(ii) Rivers
(iii) Lakes
(iv) ponds
(v) Impounding Reserviors

ponds and Lakes :- In mountains at some places natural basics are formed with impervious bed by springs and streams are known as lakes. If the water basins are in small size it is known as ponds.

Streams and rivers :- Rivers and streams are the main sources of surface water. In summer the quality of river water is better than Monsoon because in rainy season the runoff water also carries with clay. sand & silt etc...

Impounding reserviors :-

~~The~~ In some rivers the flow becomes very small and cannot meet the requirement of hot weather. In such cases the water can be stored by constructing a bund, weir or a dam across river at such places where minimum area

land is submerged in the water and maximum quantity of water to be stored

Sub surface sources :-

1. Infiltration gallaries
2. Infiltration wells
3. Wells
4. springs

Infiltration gallaries :-

A horizontal, nearly horizontal tunnel which is constructed through water bearing strate for taping underground water near rivers, lakes or streams are called infiltration gallaries.

Infiltration wells :-

In order to obtain large quantity of water the infiltration wells are sunk in series the banks of rivers. The wells are closed at top and open at bottom.

Springs :-

Sometimes ground water reappears at the ground surface in the form of springs. springs generally supply small amount of water.

Wells :-

A well is defined as an artifical coal or pit in the ground for the purpose of taping water. In India 75 - 85 % of Indian population has to depend on wells for water supply.

Characteristics of water :-

The raw or treated water can be checked and analyzed by studying and testing their physical, chemical & biological characteristics.

physical characteristics :-

physical analysis of water is carried out in order to determine the physical characteristics of water this includes tests for determining Turbidity, colour, Taste & odour, Temperature.

Turbidity :- If a large amount of suspended matter such as clay, silt or some other finally divided organic matters present in water it will appears to be muddy (or) cloudy (or) turbid in appearence

* For drinking water permissiable limit of 5-10 Units

Colour :- Dissolved organic matter from desaying visitation or some inorganic matter such as coloured soils etc.... May in part colour to the water

* The maximum permissiable colour for domestic supplier is 20 PPM (parts per million)

Taste & odour :-

(i) The dissolved organic materials the inorganic salts or the dissolved gas may ~~important~~ import taste & odour to the water.

* which generally occurs together. Taste & odour may caused by the presence of dissolved